

VZ - FZ series and HS-X/E OLED Display Replacement

Open Hardware Installation Guide

Document Type:	Open Hardware Installation Guide
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License:	See the LICENSE files in the project repository

Safety Notice

This document describes installation procedures involving access to internal electronic assemblies of a mains-powered instrument. Installation must be performed only by qualified personnel familiar with electrical safety procedures.

This project is an open hardware display replacement design for repair and restoration purposes. It is not a finished consumer product, not a CE-marked commercial kit, and not intended for standalone operation.

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This documentation and the associated design files are released under the license terms stated in the project repository. Redistribution, modification, fabrication, and use are permitted only under those license terms.

The documentation is provided as-is and may change without notice.

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1 Regulatory and Safety Information

1.1 EMI and EMC Considerations

The installation of additional electronic modules may alter the original electromagnetic characteristics of the instrument.

The installer is responsible for ensuring that:

- internal wiring is kept short and properly routed
- ground connections are secure
- no unintended coupling into audio paths occurs
- the modified instrument continues to operate safely and without excessive emissions

Improper installation may increase susceptibility to electromagnetic interference or introduce noise into the audio signal path.

1.2 Regulatory Compliance Notice

The open hardware design files and this installation guide are not a finished commercial electronic product.

The regulatory status of any assembled board or modified instrument may differ from the original factory configuration of the host instrument.

Any modification of mains-powered equipment may affect compliance with:

- EMC regulations
- electrical safety requirements
- regional product safety standards
- local repair, resale, and consumer protection rules

Anyone who fabricates, assembles, imports, sells, gives away, distributes, installs, or otherwise supplies hardware based on this project must determine which legal and regulatory obligations apply to that activity.

The author cannot control third-party fabrication, assembly quality, installation conditions, or later distribution of hardware based on these files.

1.3 CE Marking Status

No CE marking is claimed for this open hardware project, this documentation, the firmware source code, or the design files.

No CE marking should be applied to assembled hardware based on this project unless the responsible person or organization has completed the required conformity assessment, prepared the necessary technical documentation, and accepted responsibility for the resulting hardware and its intended use.

The display replacement design is not intended to be placed into service as a standalone device and has no functional use outside the designated host instrument.

The regulatory conformity of the complete instrument after modification remains the responsibility of the installer, equipment owner, or any party that makes assembled hardware available to others.

1.4 No Warranty and No Guaranteed Support

This project is provided as-is. No warranty, guarantee, or promise of fitness for a particular purpose is given, to the maximum extent permitted by applicable law.

The author may publish updates, corrections, or improvements, but there is no obligation to provide maintenance, installation support, replacement parts, or future firmware updates.

1.5 Electrical Safety Warning

Although the replacement display itself operates at low voltage (+5 V), installation requires opening the synthesizer and accessing the internal power supply unit.

The instrument contains hazardous mains voltages (230 V AC in Europe, 110 V AC in the USA and other regions). Contact with these voltages can cause severe injury, electric shock, or death. Capacitors inside the power supply may remain charged even after the unit has been disconnected from the mains.

Caution: Risk of electric shock. Disconnect the instrument from mains power and allow sufficient time for internal capacitors to discharge before opening the unit.

1.6 Qualified Personnel Requirement

All installation and modification work described in this document must be performed by qualified technical personnel with appropriate experience in electronic equipment and mains-powered devices.

This project is not intended for installation by untrained users.

1.7 Liability Disclaimer

This project is provided as-is, without any express or implied warranty or guarantee, to the maximum extent permitted by applicable law. The installation procedures described herein involve modifications to internal electronics and are performed entirely at the installer's own risk.

The author and project maintainers assume no responsibility or liability for:

- personal injury or death
- damage to the instrument
- damage to other equipment or property
- improper installation or misuse
- modifications performed contrary to this documentation
- regulatory non-compliance of assembled hardware or modified instruments

No guarantee is given for suitability, long-term reliability, regulatory conformity, or compatibility with modified instruments. By fabricating, assembling, installing, or using hardware based on this project, the user acknowledges these risks and accepts full responsibility for the outcome of the modification. This document is technical guidance only. It is not legal, regulatory, or professional safety advice.

1.8 Open Source Project Notice

The author does not offer this project as a regular commercial product, retail spare part, CE-marked kit, or installation service. Images of assembled boards and installation examples are shown for documentation purposes only and do not constitute an offer to sell.

Publication of design files, firmware, or documentation does not imply that any assembled hardware is certified, approved, CE marked, or suitable for every instrument condition.

This project:

- is not a finished consumer product
- is not offered as a regular retail product or commercial spare part
- is not a CE-marked standalone device or CE-marked kit
- is not intended for independent operation outside the host instrument
- has no functional use outside the designated host instrument
- is intended for technically qualified builders and installers

The author does not provide a commercial installation service and does not assume responsibility for hardware fabricated, assembled, sold, modified, or installed by third parties.

Anyone who manufactures, imports, sells, gives away, distributes, or installs assembled hardware based on this project is responsible for checking and complying with all applicable local safety, EMC, CE, tax, consumer protection, and product liability requirements.

The installer is responsible for ensuring that the modified instrument remains safe and suitable for use after modification.

1.9 General Recommendation

If you are uncertain about any step of the installation process, do not proceed. Consult a qualified service technician.

2 Document Purpose and Scope

This document describes how to install an OLED display module built according to the VZ - FZ - HS series OLED Display Replacement Open Hardware Project into compatible Casio VZ, Casio FZ and Hohner HS series instruments. It is intended for qualified technical personnel and experienced hobbyists with practical knowledge of electronics, mains-powered equipment, soldering, ESD precautions, and delicate connectors.

The design files, firmware, and documentation are published for repair, restoration, research, and hobbyist use.

3 Firmware Information

The display replacement module operates with firmware developed specifically for this hardware platform.

The firmware controls display initialization, timing, and communication with the synthesizer mainboard. No user configuration is required for normal operation after a working firmware version has been flashed.

Firmware source code and, where useful, prebuilt binary release files may be published through the project repository. Firmware may be improved over time to enhance stability, compatibility, or functionality.

Note: Firmware updates are optional and should only be installed if they are explicitly recommended in the release notes. If the display operates normally, no firmware update is required.

For update instructions and the latest information, refer to the project repository.

3.1 GitHub Repository and Firmware Updates

The project repository contains or may contain the hardware design files, firmware source code, firmware binaries, release information, update notes, bill of materials, PCB manufacturing files, and documentation required to build or service the display replacement design.

https://github.com/DigitalArtDeco/FW_VZ_FZ_HS-XE

Future firmware updates may also be made available through this repository when required. These updates may include improvements related to display timing, compatibility, stability, or additional functionality.

Note: Firmware updates should only be installed if they are explicitly recommended in the release notes. If the display operates normally, no firmware update is required.

Before flashing a firmware file, always verify that the binary is intended for the correct hardware revision and supported instrument model. Using an incorrect firmware file may prevent the display replacement module from operating properly.

Caution: Do not interrupt power during flashing. A failed or interrupted firmware update may leave the display replacement module in a non functional state.

4 Intended Models and Compatibility

This replacement display PCB is designed specifically for the Casio VZ-1, VZ-10M, FZ-10M, FZ-20M and their siblings Hohner HS-1/E and HS-2/E. The mechanical form factor and connector layout of the replacement display matches the original compatible display configuration and fits directly into the corresponding mainboard connector.

Note: The Casio FZ-1 uses a different display form factor and a different mainboard connector. This replacement display module is therefore NOT compatible with the FZ-1. For this reason, the model in question will not be mentioned again in this document.

4.1 Model Specific Installation Overview

The following table summarizes the model specific installation requirements. Always verify the actual PSU and mechanical layout before performing any modification, especially on instruments that may have been modified previously.

Model	7805 regulator upgrade	Dedicated PSU 5 V wiring	Internal metal frame removal	VZ-1 display mounting frame
Casio VZ-1	Required	Required	Not required	Must be reused from the original display
Casio VZ-10M	Required	Required	Required. Internal metal frame behind the display area must be removed.	Not used
Casio FZ-10M	Not required	Not required	Normally no metal frame installed	Not used
Casio FZ-20M	Not required	Not required	Normally no metal frame installed	Not used
Hohner HS-1/E	Not required	Not required	Normally no metal frame installed	Not used
Hohner HS-2/E	Required	Required	Required. Internal metal frame behind the display area must be removed.	Not used

Table 1: Model specific installation requirements for the OLED display replacement.

Before fabricating, assembling, or installing, users should:

- Verify the model designation of their instrument
- Compare the original display connector with the adapter PCB
- Confirm mechanical clearance and mounting hole alignment

If there is uncertainty regarding compatibility, inspection of the instrument before installation is strongly recommended.

Note: Instruments that had been already modified by previous owners may require additional adjustments. Please check these before proceeding.

5 Builder and Installer Expectation Checklist

This checklist summarizes normal characteristics of the open hardware display replacement design and its installation. These items are expected behavior and do not indicate a fault.

5.1 Display Characteristics

- The OLED screen size is 2.42 inch and therefore slightly smaller than the original display. That is, small margins around the visible screen area are normal.
- The new display uses modern OLED technology instead of the original LCD display type with an electro-luminescence backlight.
- That's why the brightness and, in particular, the contrast are significantly better than those of the original display.
- The display's lifespan also increases more than tenfold, from max. 2000 hours with LED backlight to expected 20.000 hours with OLED.



Figure 1: Replacement display installed in the VZ-1. Notice a slightly smaller size.

5.2 Installation Characteristics

- The display connector adapter PCB may require slightly more insertion force than the original flex cable.
- For VZ-1, VZ-10M and HS-2/E units, the replacement module requires a dedicated +5 V supply from the PSU and therefore an upgraded 7805 voltage regulator.
- The original display metal frame in all rack versions will be removed during the installation.

5.3 Required Hardware Assemblies

Depending on the chosen build method, the following hardware assemblies are required. They are listed here just as technical installation elements and not as a commercial package or sales offer.

- OLED display replacement module assembly built from the project files
- Connector adapter PCB built from the project files
- XH wire or equivalent cable to connect the display module with the connector adapter PCB
- A 7805 voltage regulator, depending on which model is being upgraded

5.4 Required Tools

- Phillips screwdriver
- Soldering iron and solder
- Flashlight or work light
- ESD protection, wrist strap or ESD mat recommended
- We also recommend using a desoldering tool to gently replace individual original components as needed

6 Safety and Disclaimer to the Replacement Procedure

Caution: Disconnect power before opening the instrument. Unplug the mains cable and wait at least 5 minutes before touching internal electronics.

Note: ESD: The replacement module and instrument electronics can be damaged by static discharge. Proper grounding is strongly recommended.

Note: Do not force anything. If a part does not fit, re-check connector orientation and mechanical alignment.

7 Module Overview and Connectors

7.1 Front side of the replacement PCB

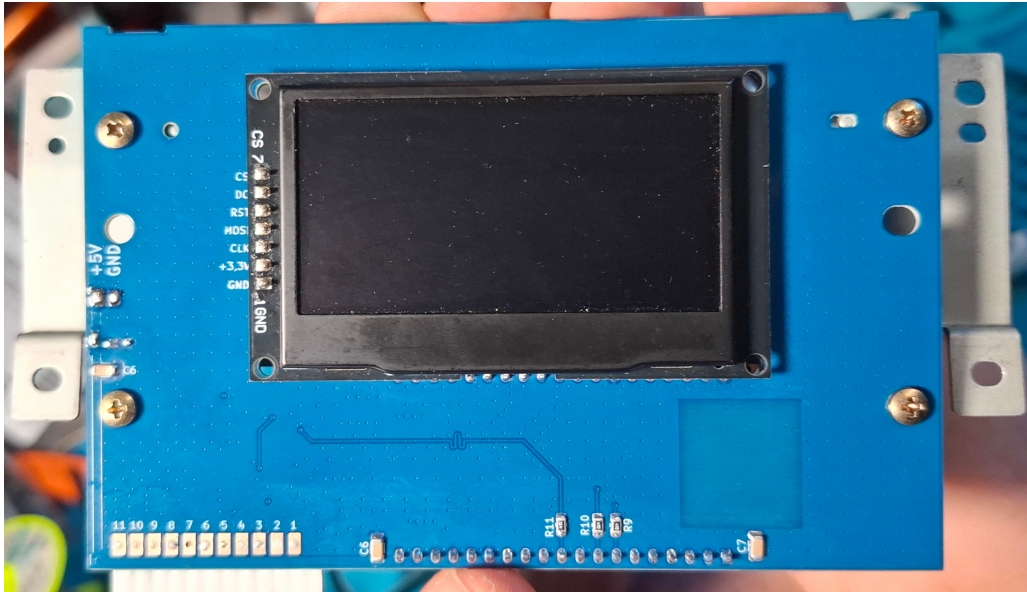


Figure 2: Replacement module, top view.

7.2 Rear side of the replacement PCB

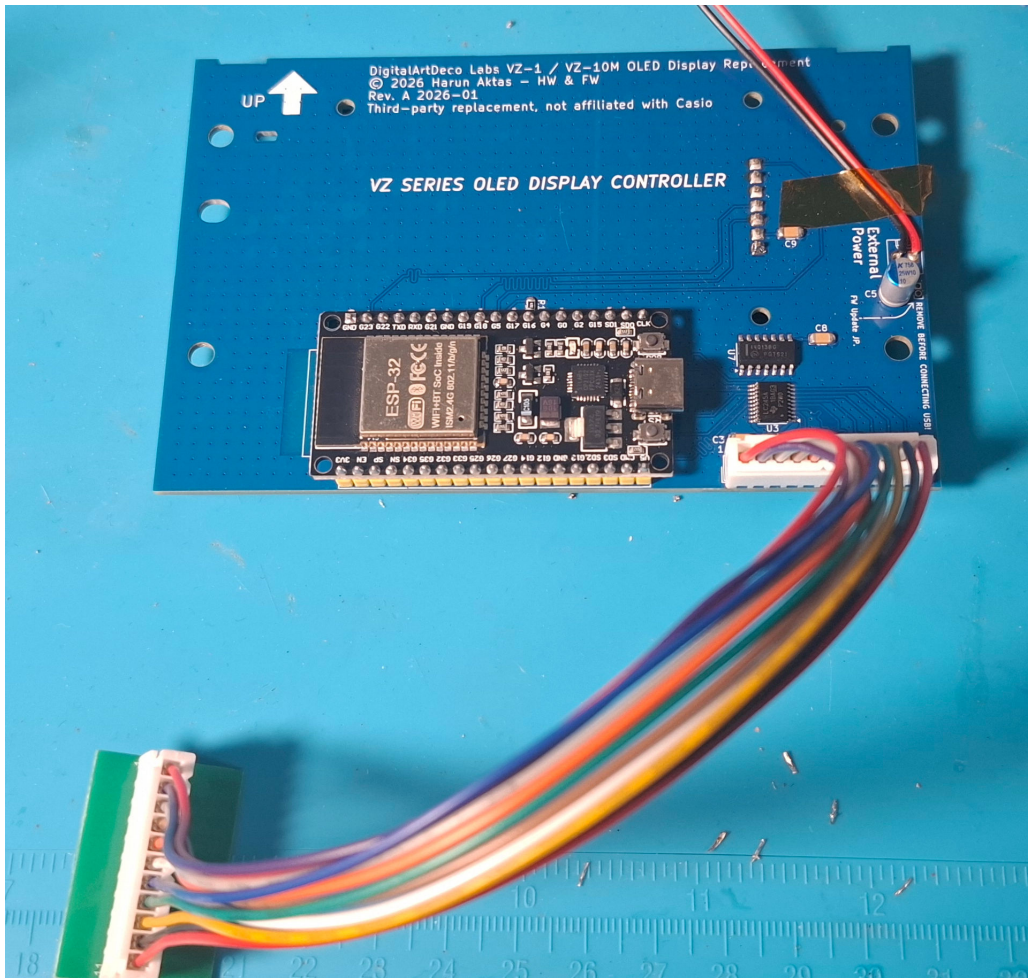


Figure 3: Replacement module with rear view showing the XH connector on the PCB and the connection wire with the adapter PCB.

8 Preparation

Set up your workspace. This will help you stay organized:

- Clean, well-lit workspace
- Small tray for screws and parts, labelled if needed
- Take photos during disassembly to support correct reassembly

9 Removing the Original Display

Caution: Your device is at least 35 years old. Be careful with fragile connectors and aged plastic parts. Avoid bending or twisting ribbon cables.

1. Locate the original display assembly.
2. Disconnect the display ribbon cable from the motherboard by pulling it out.
3. Remove mounting screws or brackets holding the display.
4. Lift the display assembly out gently.
5. Locate the four screws securing the metal frame to the original display PCB.
6. Remove all four screws.
7. Carefully separate the metal frame from the original display.
8. Retain the frame and screws in case to reuse them.

The metal mounting frame in the original display assembly of the Casio VZ-1 positions the display correctly inside the instrument and provides the mechanical mounting interface. For the replacement display in the VZ-1 this metal frame must be reused.

The rack models like the VZ-10M or HS-2/E also use a mounting frame for their original display. This metal frame **MUST NOT** be used together with the new display module. Otherwise, there would be a risk of a short circuit on the new display module. The replacement PCB is designed with the required mounting holes and clearances to fit these units directly. This makes the metal frame unnecessary.

10 Installing the Replacement Display

1. Place the replacement module in the original display position.
2. Align mounting holes and standoffs.
3. Install screws lightly first, then tighten evenly.

10.1 Internal Metal Frame Removal for VZ-10M and HS-2/E



Figure 4: Internal metal frame in the VZ-10M and HS-2/E display area. Remove this frame before installing the replacement display to prevent possible short circuits.

10.2 Re-installing the metal frame on the replacement PCB for the VZ-1

1. Align the metal frame with the mounting holes on the replacement PCB.
2. Ensure the orientation matches the original assembly.
3. Reinstall the four screws using the original hardware.
4. Tighten evenly without excessive force.

The replacement PCB includes matching mounting holes and cutouts to accept the original VZ-1 metal frame.

10.3 Illustration for Casio VZ-1

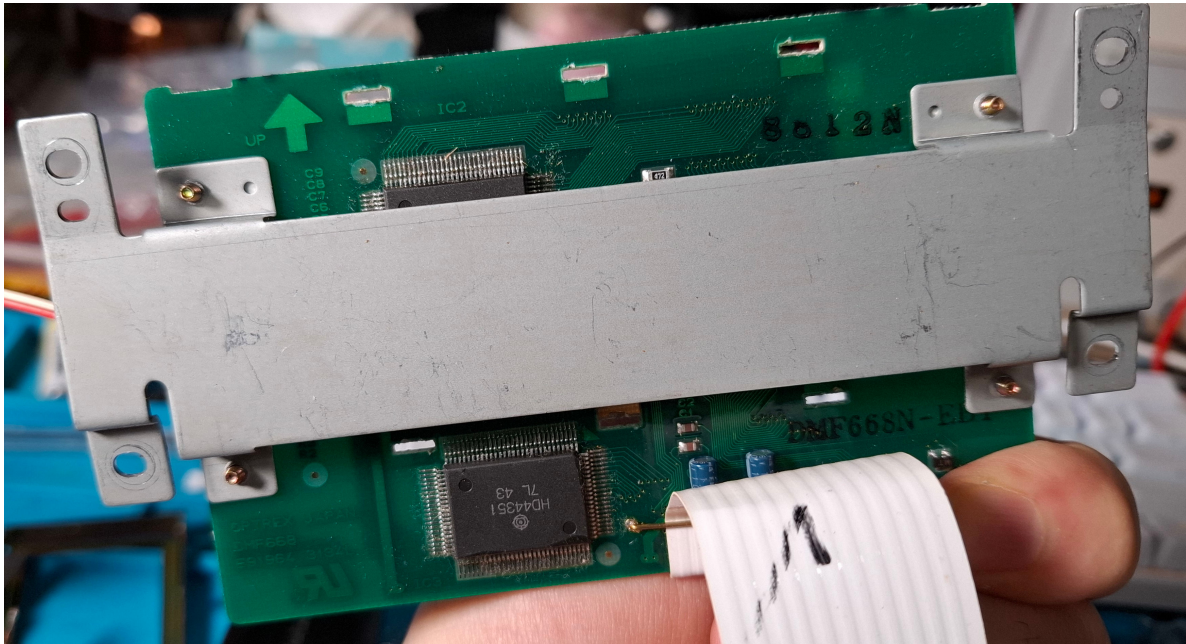


Figure 5: Original VZ-1 metal mounting frame on the old display.



Figure 6: Original metal mounting frame on the new display.

11 Power Supply Modification (Model Dependent)

11.1 Background

The replacement OLED module requires approximately 100 mA additional current at +5 V. For the Casio VZ-1, Casio VZ-10M and Hohner HS-2/E, the original 5 V power supply design has limited current reserve. Although the instrument may power up and operate initially, the original regulator may reach elevated temperatures during extended operation. Thermal stress may result in increased noise coupling into the audio path, voltage instability, or unreliable display operation. To avoid such problems from the outset, in these models the original 7805 voltage regulator is replaced with a higher current device and the replacement display will be powered directly from the power supply unit.

Note: The 7805 voltage regulator upgrade is required only for VZ-1, VZ-10M and HS-2/E. It is not required for FZ-10M, FZ-20M or HS-1/E.

11.2 New Regulator Requirements

For VZ-1, VZ-10M and HS-2/E installations, the replacement regulator must meet the following criteria:

- Fixed 5 V output
- Minimum 1.5 A continuous current rating
- TO-220 package, pin-compatible with the 7805 footprint
- Proper heatsink mounting capability

11.3 Thermal Considerations

Linear regulators dissipate power according to:

$$P_{loss} = (V_{in} - 5 V) \times I_{load}$$

Depending on the PSU input voltage and total system current, significant heat may be generated. The replacement regulator must:

- Be firmly mounted to the existing heatsink
- Use proper thermal compound if originally present
- Have adequate airflow inside the enclosure

11.4 Voltage Regulator Replacement Procedure

1. Disconnect mains power and allow capacitors to discharge.
2. Locate the original 7805 regulator on the PSU board.
3. Desolder and remove the regulator.
4. Install the higher-current 5 V regulator.
5. Mount securely to the heatsink.
6. Use thermal grease to ensure proper heat dissipation.
7. Inspect solder joints carefully.

11.5 Adding the +5 V Supply for the new Display Module

The power cable used for this installation must be soldered directly to the output of the +5 V linear regulator on the PSU board.

- Red wire → +5 V output of the 7805 regulator
- Black wire → GND

1. Locate the 7805 regulator mounted on the heatsink.
2. Identify the +5 V output pin and a nearby ground point.
3. Solder the red wire to +5 V and the black wire to GND.
4. Route the cable cleanly to the display module and secure it to prevent strain.

This creates a stable and independent supply directly from the PSU and avoids loading sensitive mainboard traces.

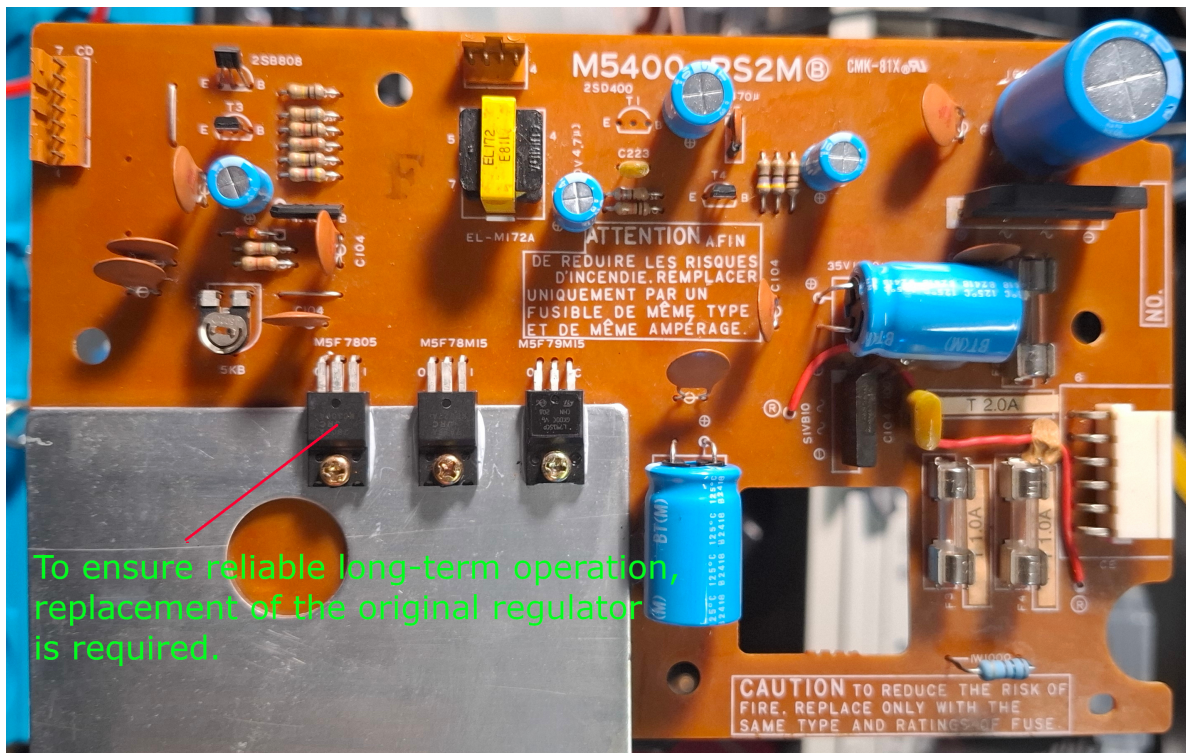


Figure 7: Location of the original 7805 regulator and replacement with higher-current device. Replace with annotated photo.

11.6 Voltage Regulator Verification

After replacing the voltage regulator, it is recommended that you check its output voltage immediately to ensure it is correct. You can do this by measuring the voltage directly at the voltage regulator, without reconnecting the connector to the synthesizer's main board just yet. The power supply will then have no load. So if you measure a voltage slightly more than 5.00V up to about 5.1V, that is within the correct range:

- **Before** connecting to the synthesizer, measure the 5 V rail under **no load**.
- Confirm voltage stability.

11.7 Disabling the High-Voltage Inverter, Transistor T1 / T18

After removing the original display, the high-voltage inverter section for its electro-luminescence back-light is no longer be required. Moreover this circuit often causes audible coil noise and consumes power unnecessarily. For this reason, the circuit is taken out of service. To do this, simply remove one transistor.

1. Locate transistor T1 for Casio VZ series or Hohner HS-2/E on the PSU board.
2. Locate transistor T18 for FZ series or Hohner HS-1/E
3. Desolder and remove T1 or T18 without replacing it.

Removing T1 or T18 disables the inverter drive stage and prevents unnecessary power consumption and acoustic noise. This modification does not affect normal operation of the replacement display.

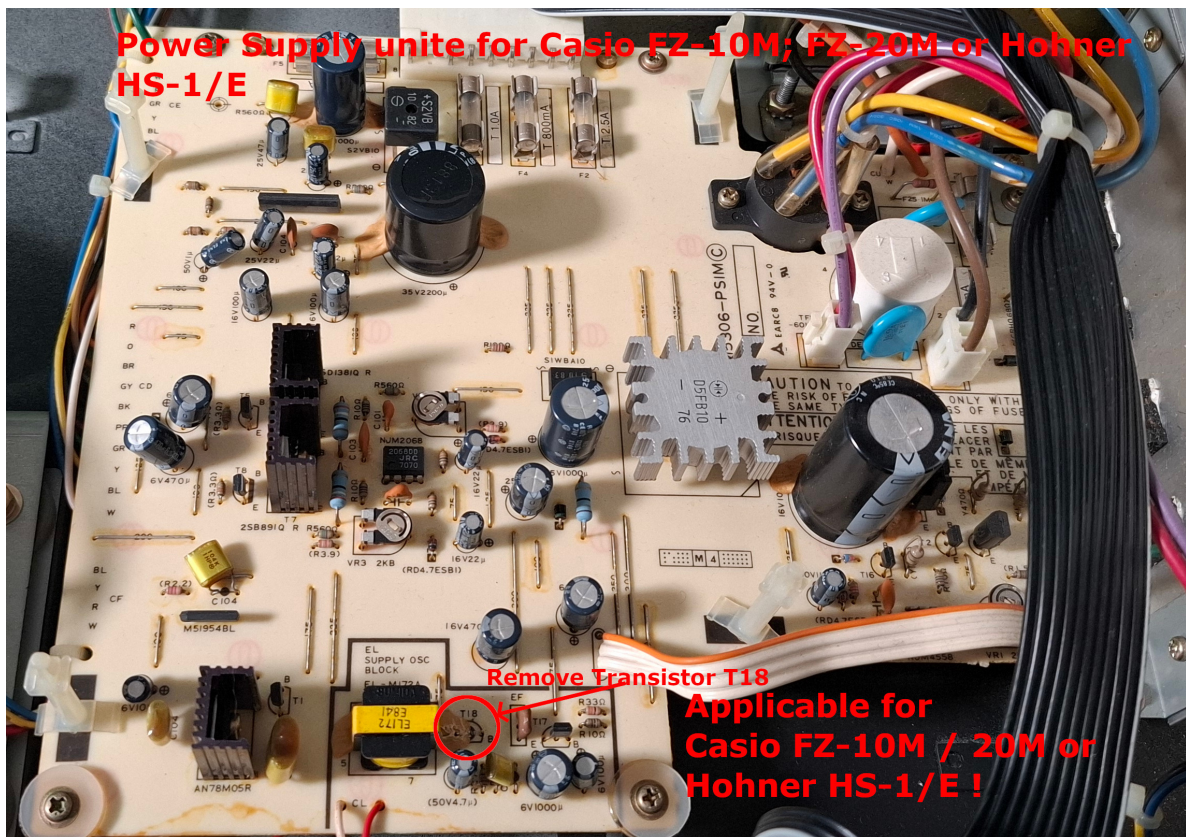


Figure 8: Location of transistor T18 for FZ-10M, FZ-20M and Hohner HS-1/E. To disable the high-voltage circuit, remove transistor T18.

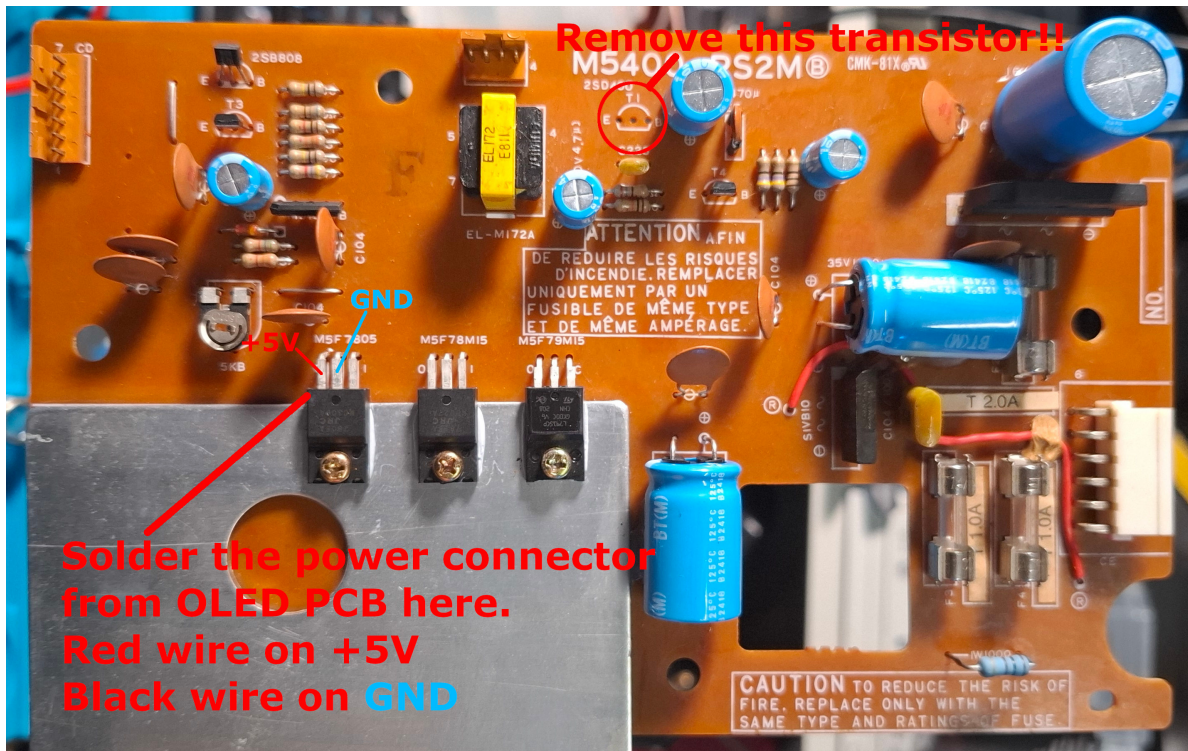


Figure 9: PSU of the Casio VZ series. In this picture the transistor T1 is already removed.

11.8 Final Inspection

Before reassembly:

- Check all solder joints for shorts or cold joints
- Ensure wires cannot touch heatsinks or sharp edges
- Verify correct polarity, +5 V and GND
- Secure the cable with tape or tie-downs

Caution: Reversing polarity or shorting the regulator output may permanently damage the display module. Double-check wiring before powering the instrument.

12 Connector Adapter PCB Installation

The adapter PCB provides the electrical and mechanical interface between the replacement display module and the original synthesizer mainboard connector. It replaces the original flexible cable connection with a rigid PCB adapter.

12.1 Mechanical Fit

The adapter PCB is designed to match the original mainboard connector geometry of the synthesizer and fits directly into the existing display socket. No modifications to the mainboard are required.

12.2 PCB Thickness

The adapter PCB thickness is approximately 0.8 mm. This is slightly thicker than the original factory flex cable. That's why it's normal for it to take significantly more force plugging in this connector than with the original plug.

12.3 Mainboard Connector Orientation

The original display connector on the synthesizer mainboard is a flex cable socket with electrical contacts located on one internal side only. For proper operation, the adapter PCB must be inserted so that the contact pads on the adapter face the same direction as the conductive contacts inside the mainboard connector.

1. Inspect the connector on the mainboard and identify the side containing the internal metal contacts.
2. Align the adapter PCB so that its contact pads face this contact side.
3. Insert the adapter straight and fully into the connector.

Note: If the adapter PCB is inserted in the wrong direction, no electrical damage will occur. The adapter is populated with contacts on one side only. In the wrong orientation, the display will simply not function because the signals are not connected.



Figure 10: Mainboard connector showing the contact side and the correct adapter PCB orientation.

12.4 Insertion Procedure

To avoid putting too much mechanical stress on the motherboard when plugging in the new connector, hold the bottom of the board steady as you insert it. If the adapter has been plugged in correctly, its contacts are almost completely inside the connector and are barely visible.

1. Align the adapter PCB with the connector before applying pressure.
2. Ensure the contact pads face the correct direction.
3. Support the mainboard from the rear side with one hand.
4. Press the adapter straight into the connector using even force.
5. Do not tilt or insert at an angle.

Caution: Do not bend the mainboard or apply pressure only from one side. Excessive force may damage the connector or PCB traces.

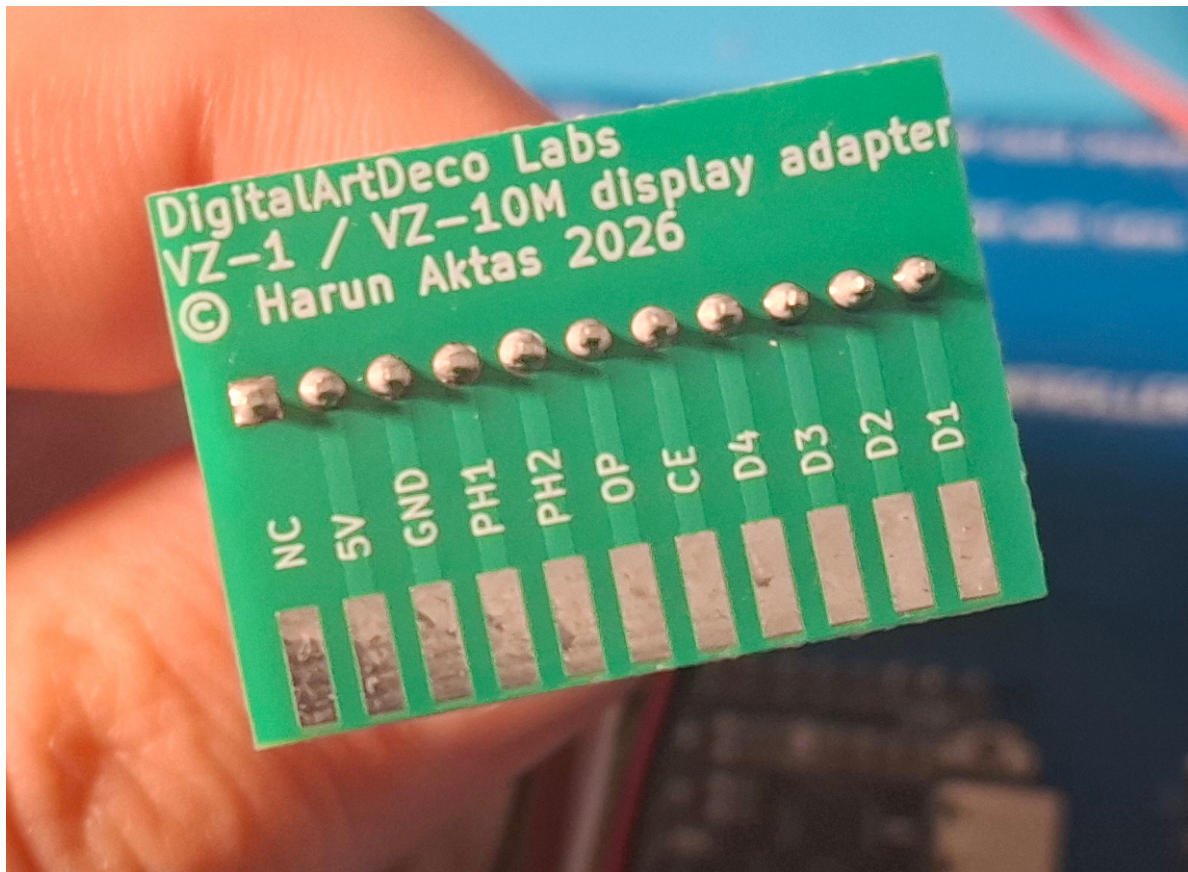


Figure 11: Adapter PCB. Assembly example for the Casio FZ-10M / FZ-20M and Hohner HS-1/E

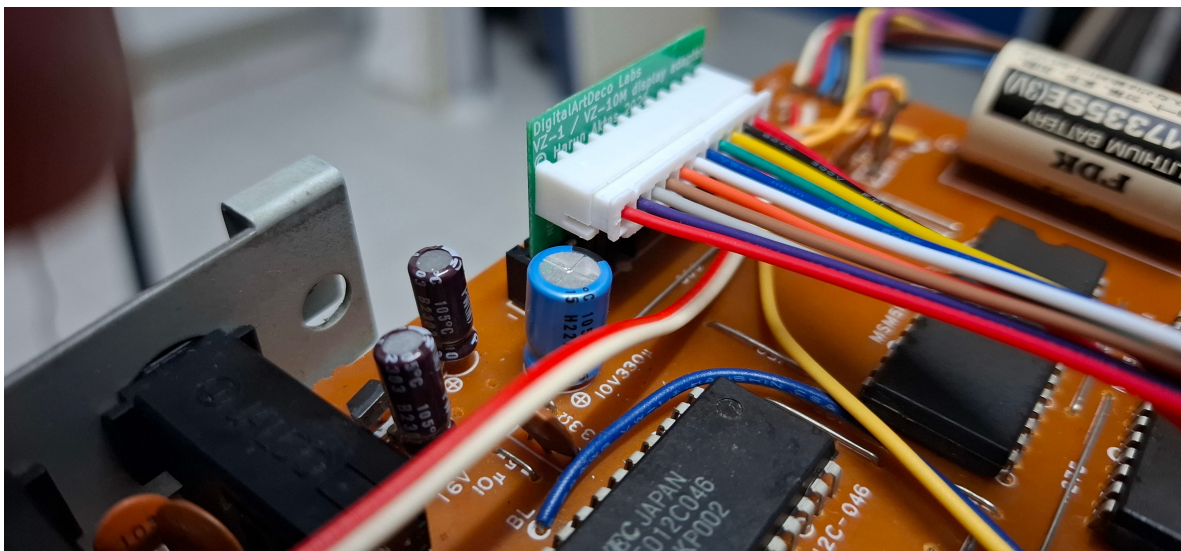


Figure 12: Adapter PCB aligned and fully inserted into the mainboard connector. Pads must be flush. This example shows the Casio version VZ-1 / VZ-10M and Hohner HS-2/E of the installation.

Note: The Casio FZ-10M / FZ-20M and Hohner HS-1/E mainboard installation differs from that of the Casio VZ-1 and VZ-10M and HS-2/E. that's why the XH connector for the VZ models and HS2/E is installed on the opposite side of the adapter PCB then in the FZ models and HS1/E.

13 Reassembly of the device

1. Inspect that no cables are pinched.
2. Reinstall covers in reverse order.
3. Reinstall all screws.

14 First Power On Check

1. Connect mains power.
2. Power on the instrument.
3. Confirm the display shows a stable image.

Caution: If you smell smoke or see unusual behavior, disconnect power immediately.

14.1 What to Expect

If the display modules is installed and powered correctly:

- The screen initializes shortly after the instrument is powered on
- Text and graphics appear stable without flickering
- Brightness is uniform across the display surface
- Characters are clearly readable from the normal viewing position

If the installed display visually matches the reference image and behaves as described above, the installation can be considered successful. Congratulations - You made it!

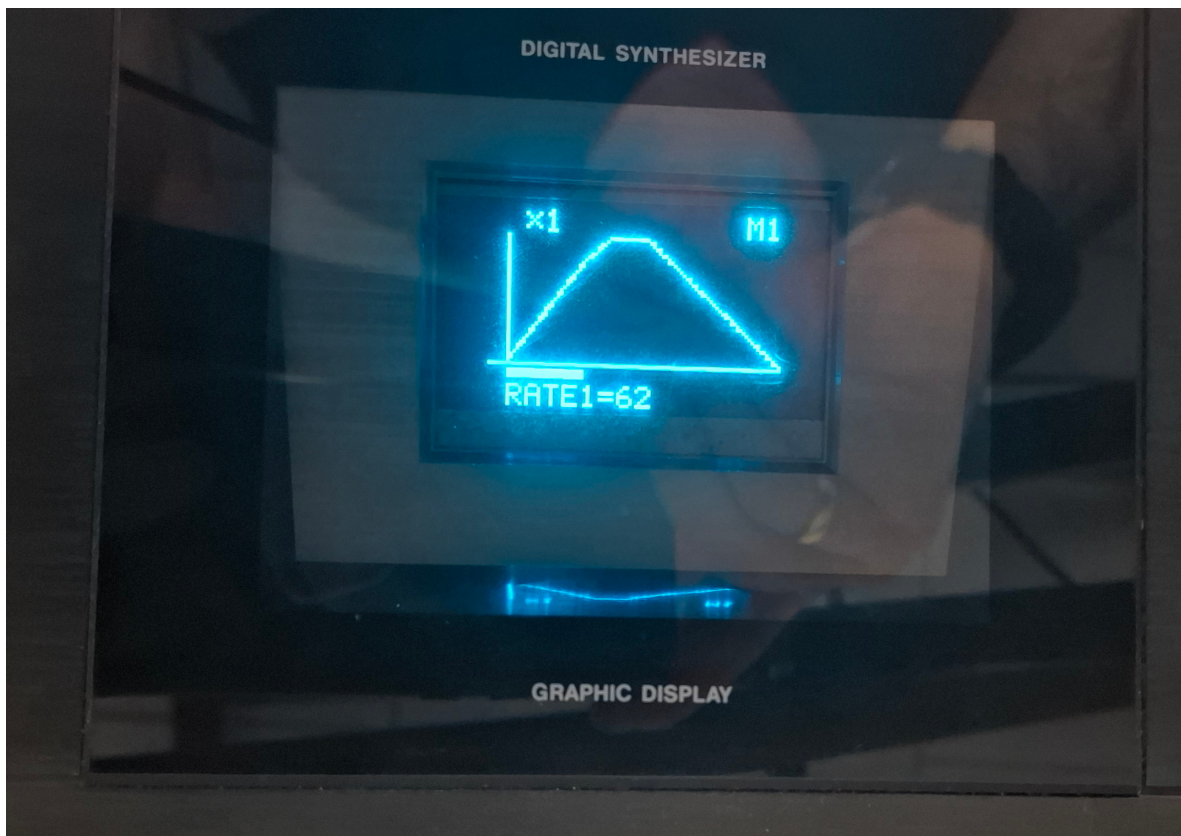


Figure 13: Replacement display installed in the instrument in normal operation mode.

15 Troubleshooting

Re-check the installation if:

- the screen remains blank
- the image flickers
- the instrument behaves unexpectedly after installation

In such cases, first verify connector seating, orientation, and power wiring before assuming a hardware defect.

Symptom	Action
No display output	Unplug, open again, re-seat the cable and check the connector orientation.
Flickering or unstable image	Check cable seating, make sure the module is mechanically secured, ensure no cable is under stress.
Display works but looks misaligned	Verify the module is seated flush and mounting points are aligned.
Audible noise after installation	For VZ-1, VZ-10M and HS-2/E, verify regulator replacement, PSU wiring and inverter disablement where applicable.

16 Contact

For questions about the open source project and the installation documentation:

- E-mail: akhar9119@gmail.com
- Project repository: https://github.com/DigitalArtDeco/FW_VZ_FZ_HS-XE
- PCB revision documented here: A

The author does not provide a commercial installation service or guaranteed support response.

17 Revision History

Version	Date	Author	Changes
1.0	June 1, 2026	<i>Harun Aktas</i>	Initial release with model-dependent 7805 regulator requirements.
1.3.0	July 1, 2026	<i>Harun Aktas</i>	Reworked wording for open source publication, non-commercial project status, liability disclaimer, and CE marking clarification.
1.4.0	July 4, 2026	<i>Christoph Oettinger</i>	Changes to the chapter sequence, repeats removed.